

06. Basic Network Implementation, Database Management and Detecting Attacks

Aim

To perform basic network checks, demonstrate simple database operations, and inspect Windows security events.

Network Information

```
Get-NetIPConfiguration
Get-NetIPAddress
Get-NetAdapter
```

Connectivity Test

```
Test-Connection google.com -Count 4
```

DNS Lookup

```
Resolve-DnsName google.com
```

TCP Connections

```
Get-NetTCPConnection
```

Simple SQLite Database Example

```
Install-Module PSSQLite -Scope CurrentUser
Import-Module PSSQLite
Invoke-SqliteQuery -DataSource "students.db" -Query "CREATE TABLE IF NOT EXISTS Students (Id
INTEGER, Name TEXT, Course TEXT)"
Invoke-SqliteQuery -DataSource "students.db" -Query "INSERT INTO Students VALUES (1, Rahul, BCA)"
Invoke-SqliteQuery -DataSource "students.db" -Query "SELECT * FROM Students"
```

Detecting Failed Windows Logons

```
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4625} -MaxEvents 20
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4625} -MaxEvents 100 | Measure-Object
```

Security Event ID 4625 can be used to demonstrate monitoring of failed logon attempts. This is a defensive monitoring exercise; it does not perform an attack.